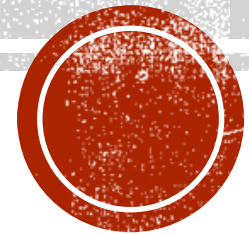


INTRODUCTION TO RECONFIGURABLE COMPUTING

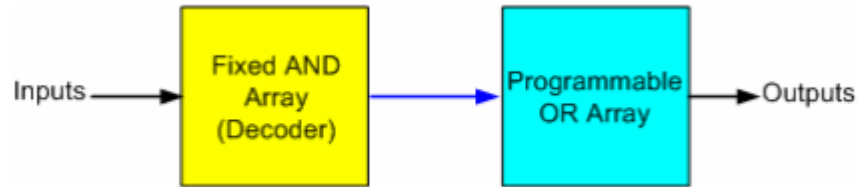
CS G553: Reconfigurable computing

Dr. Meetha V Shenoy

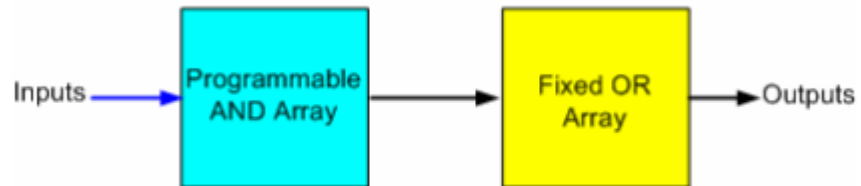
Lect-4



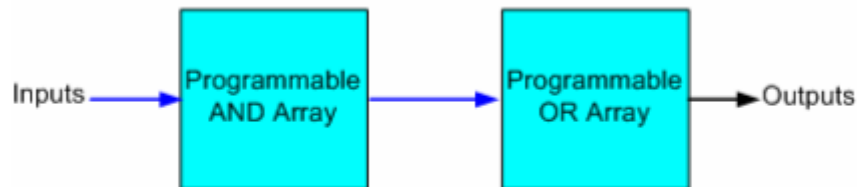
FUNDAMENTAL PLD TYPES



(a) Programmable Read Only Memory (PROM)



(b) Programmable Array Logic (PAL) Device



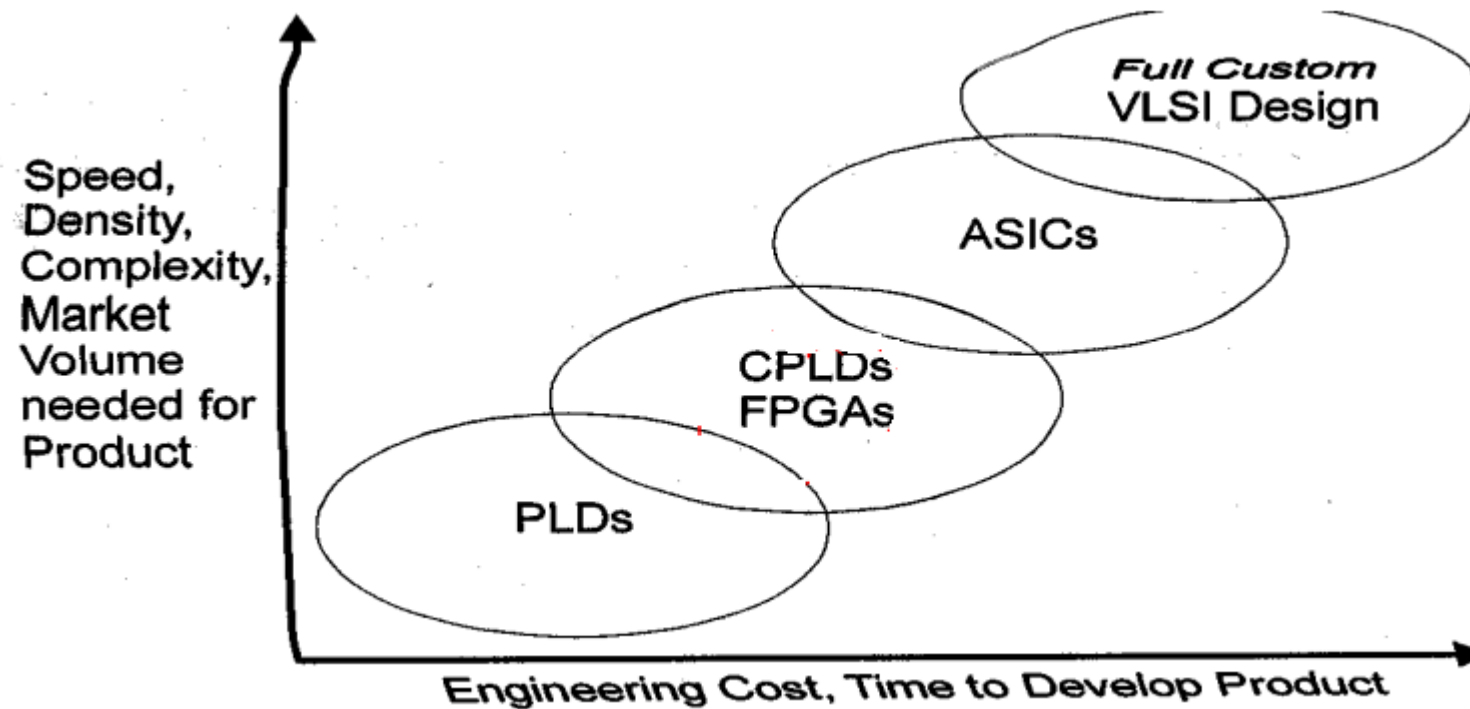
(c) Programmable Logic Array (PLA) Device

→ Programmable Connections

→ Normal Connections



DIGITAL LOGIC



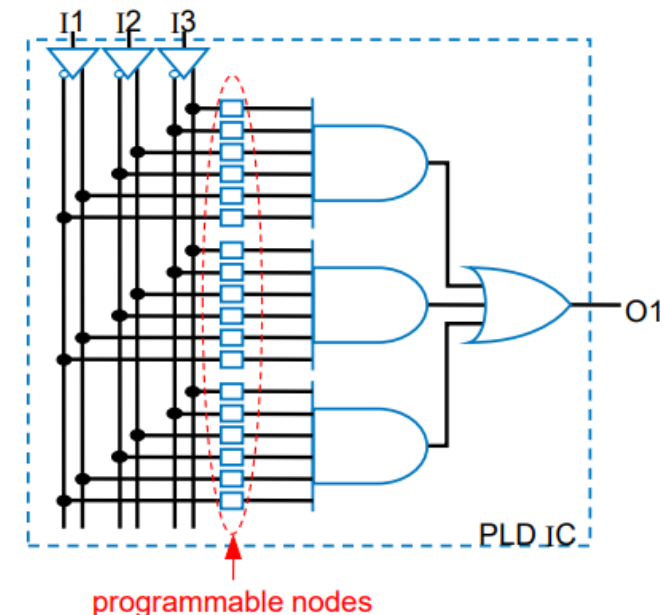
LOGIC DEVICES

- SPLD (Sequential or Simple Programmable Logic Device)
- CPLD (Complex Programmable Logic Device)
- FPGA (Field Programmable Logic Device)

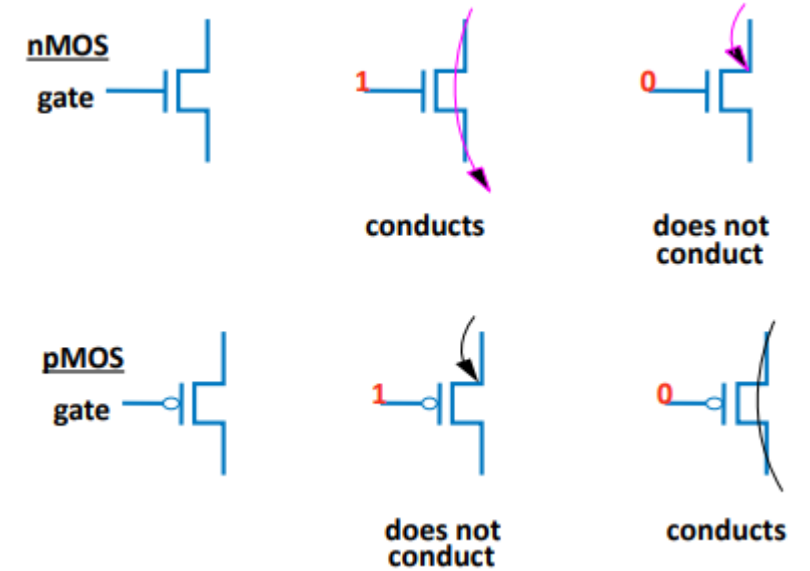
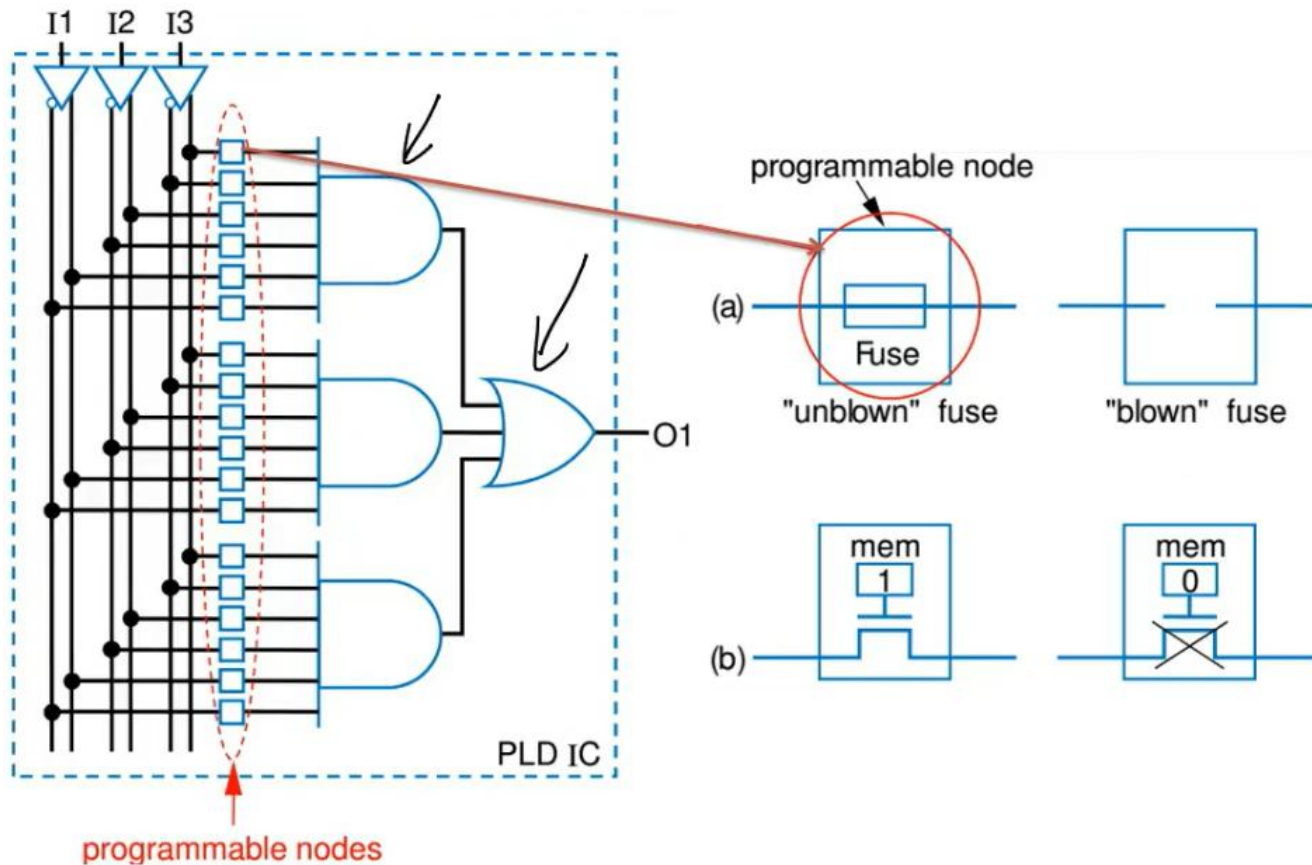


SPLD

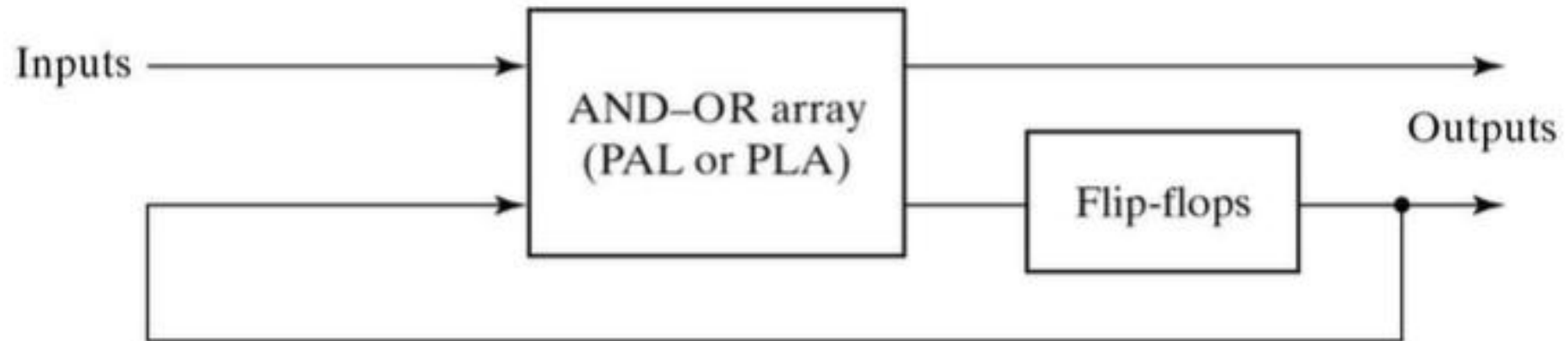
- Developed 1970s (thus, pre-dates FPGAs)
- Prefabricated IC with large AND-OR structure
- Connections can be "programmed" to create custom circuit
 - Circuit shown can implement any 3- input function of up to 3 terms



SPLD

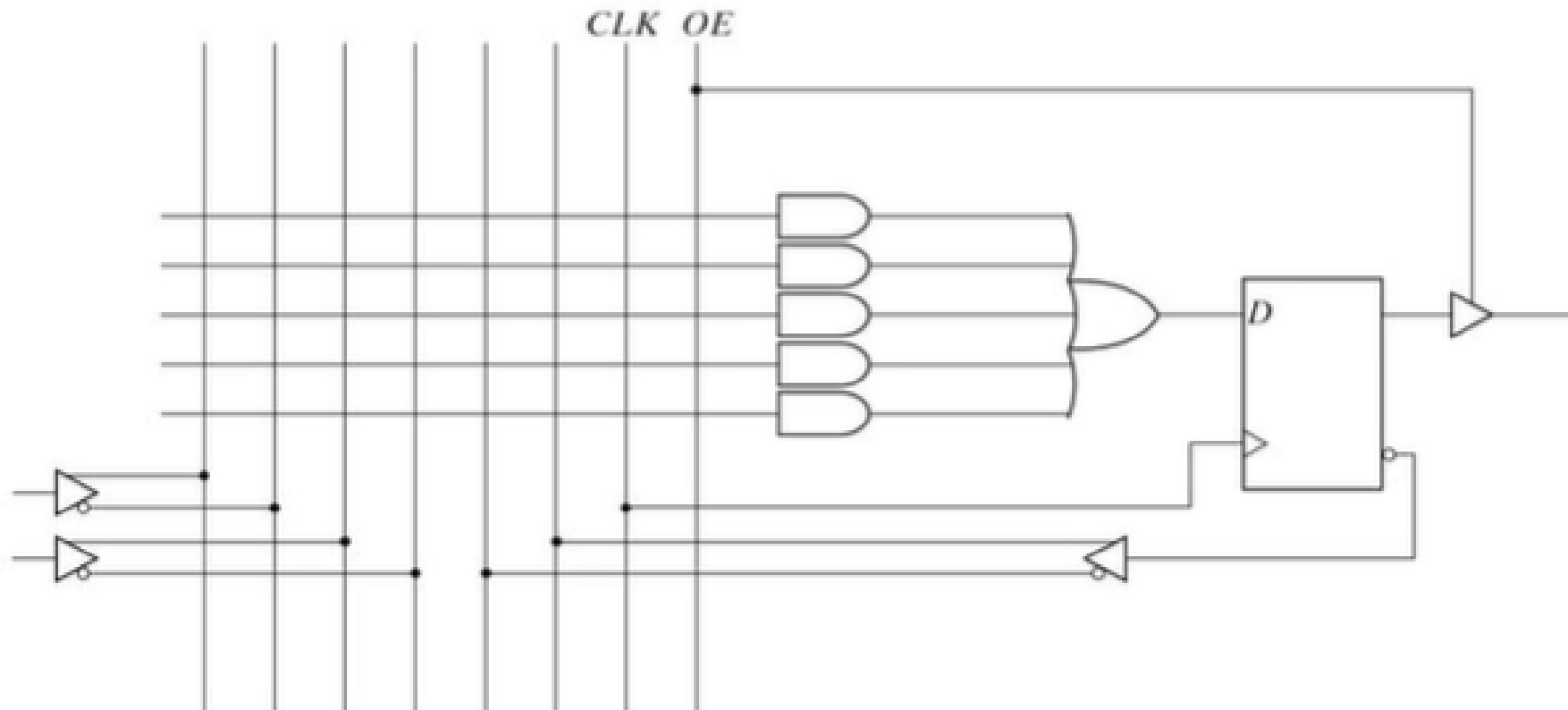


SPLD

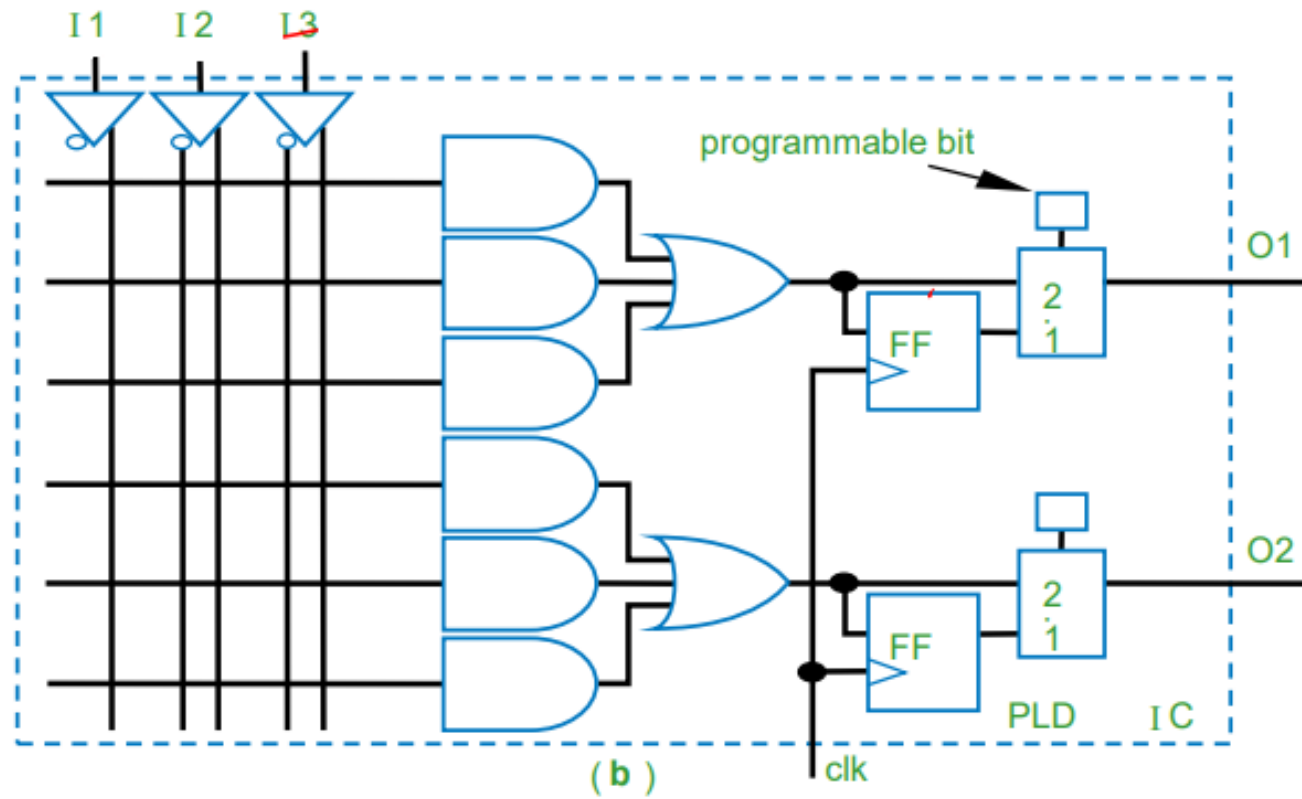


MACROCELL

- Each section of SPLD is called a macrocell.



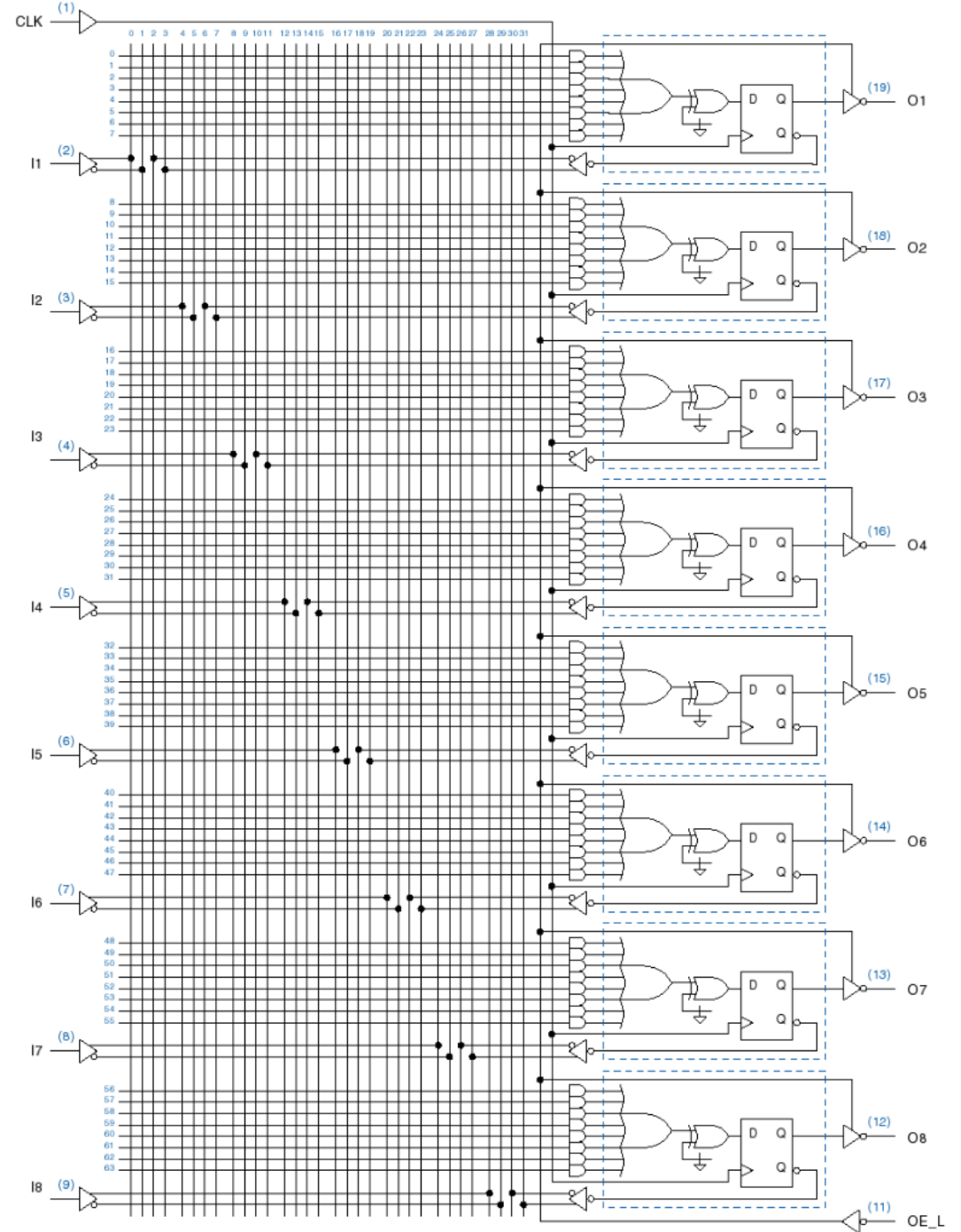
MACROCELL



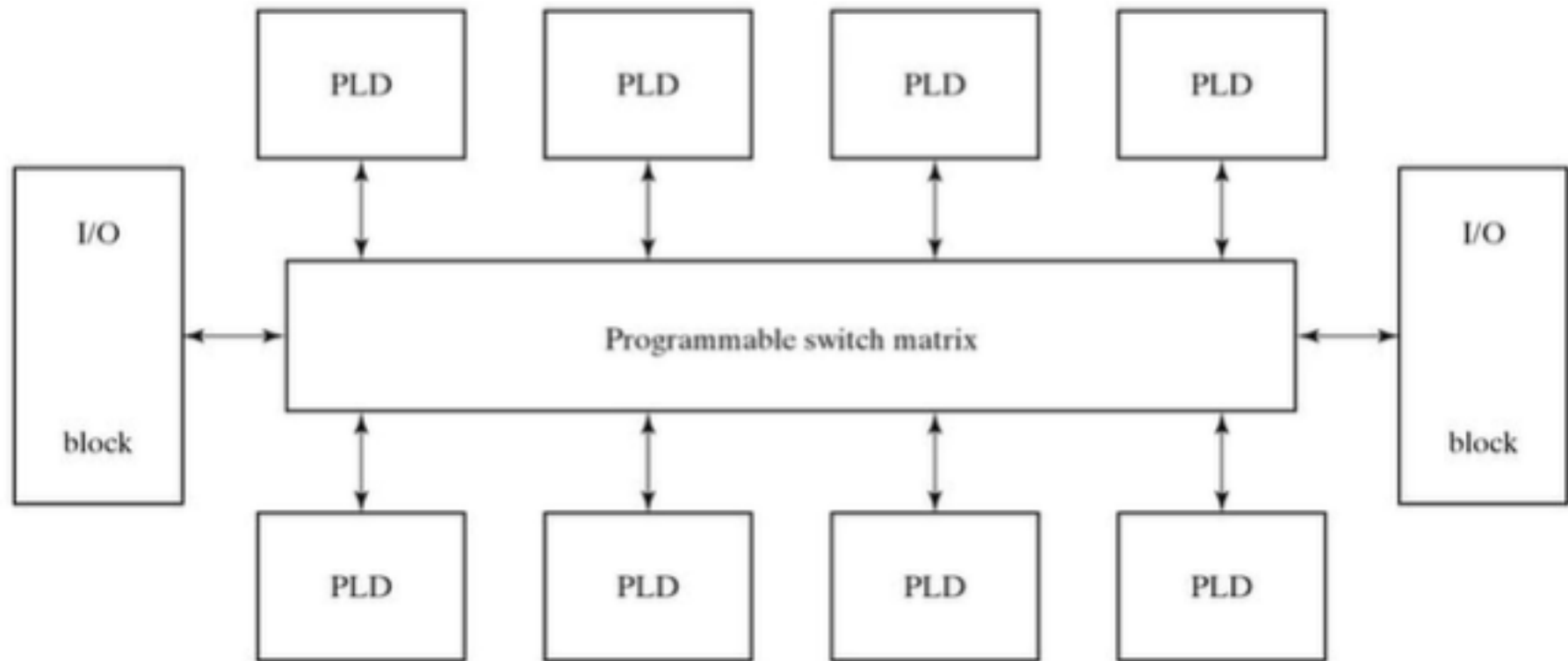
Macrocell



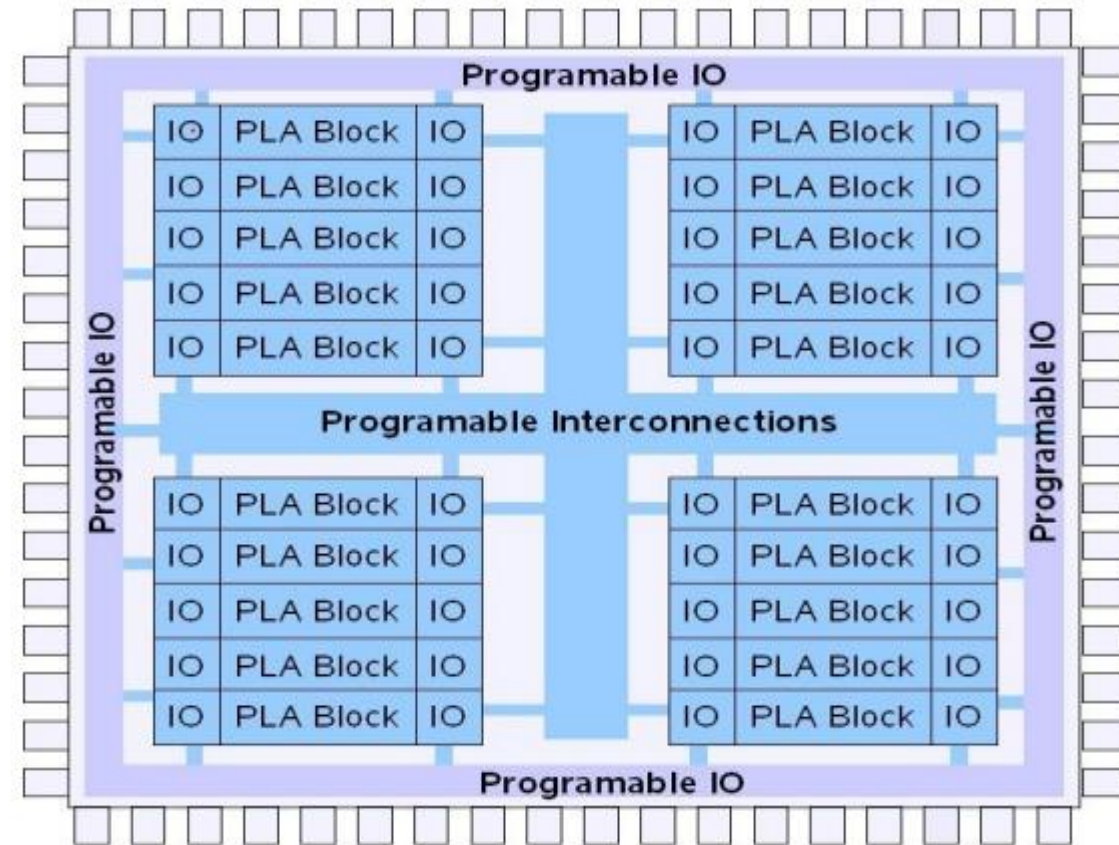
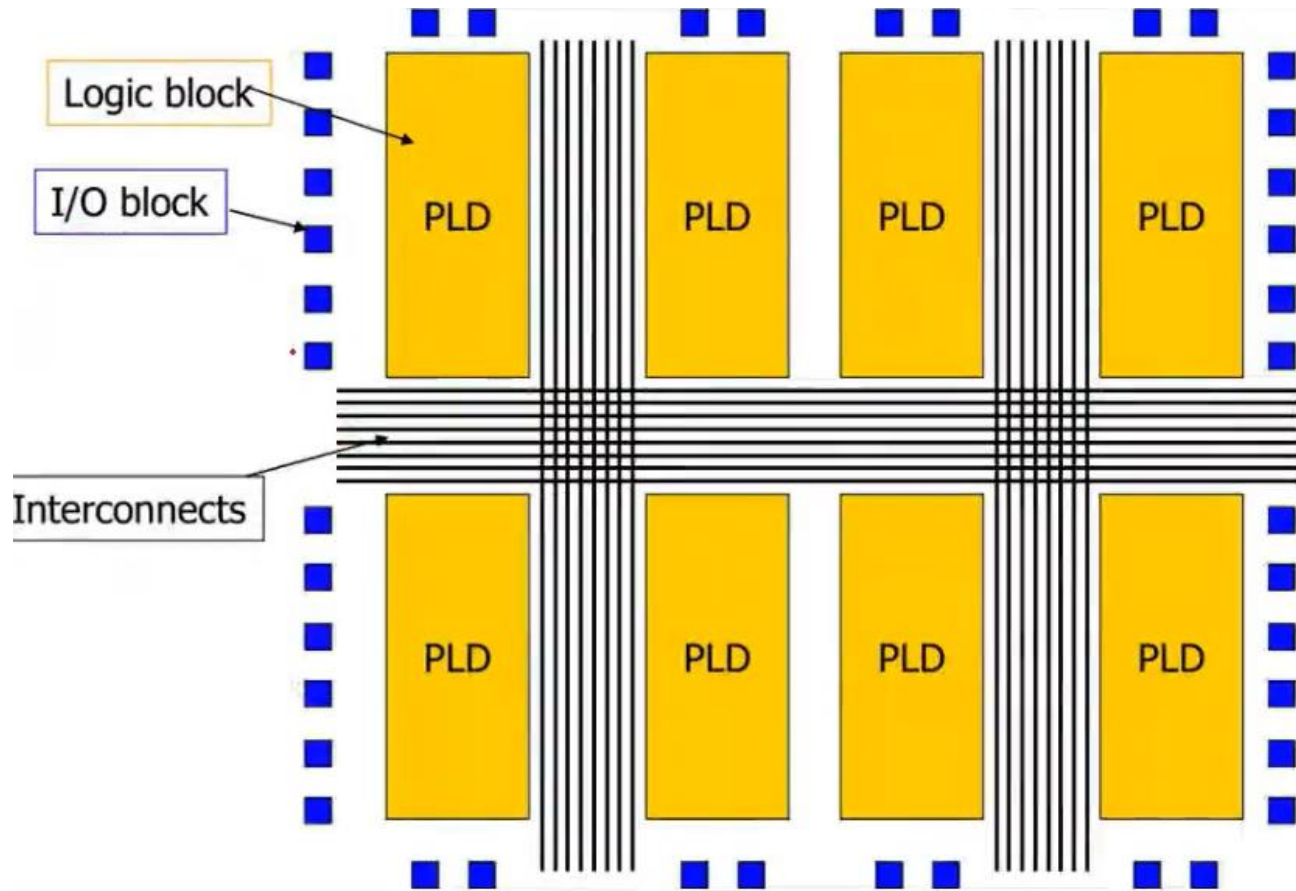
GAL16V8



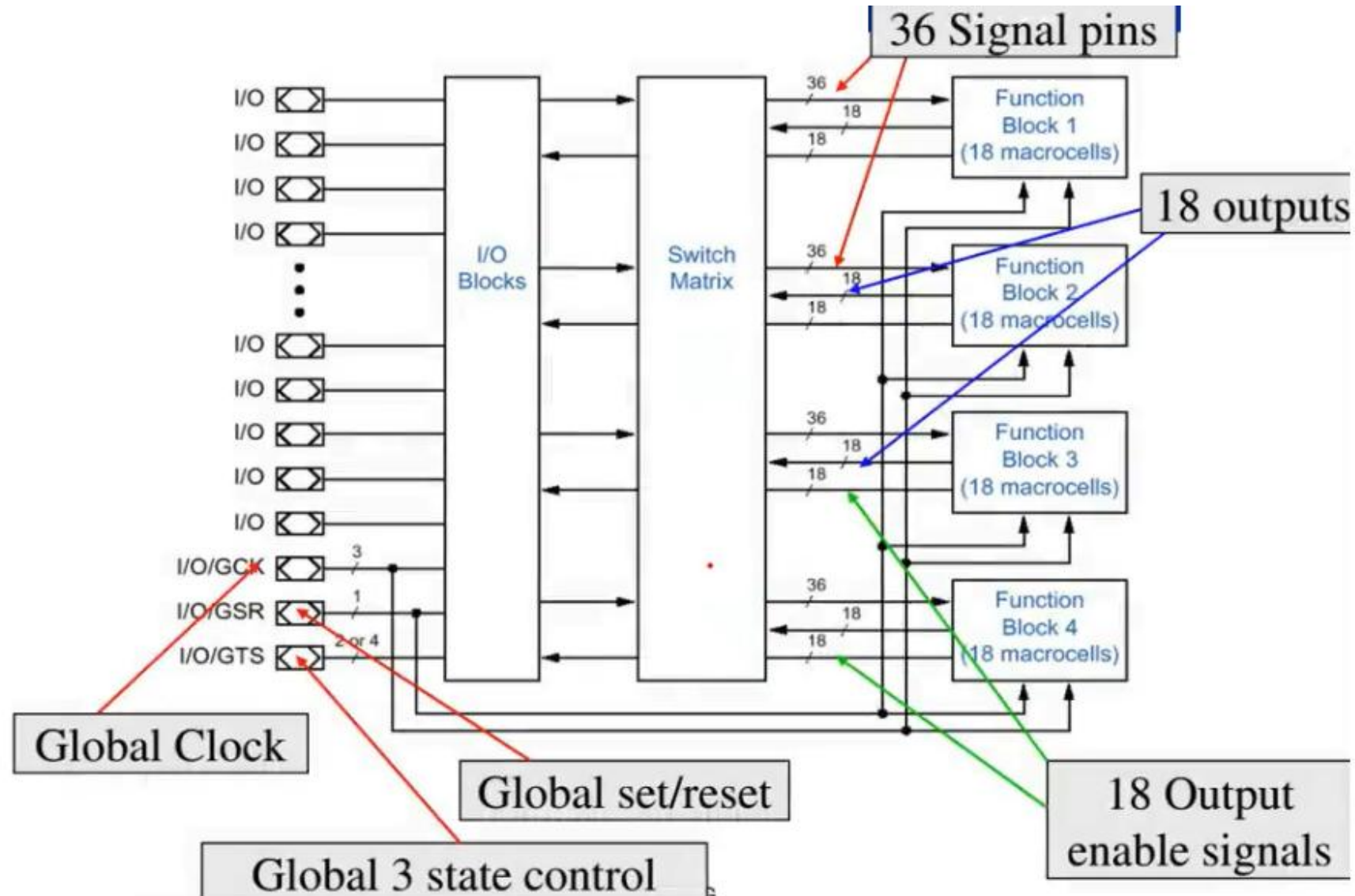
CPLD



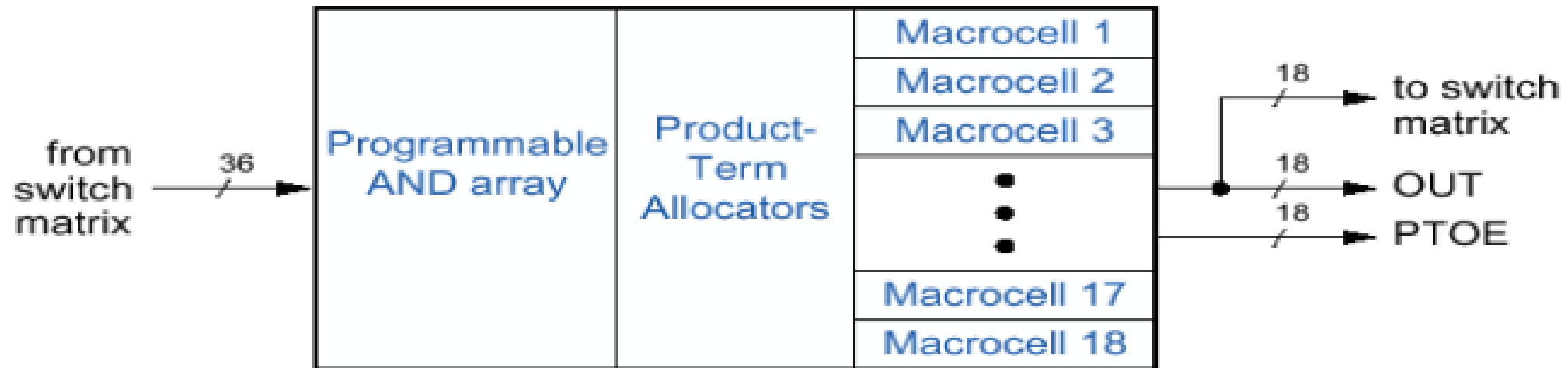
CPLD

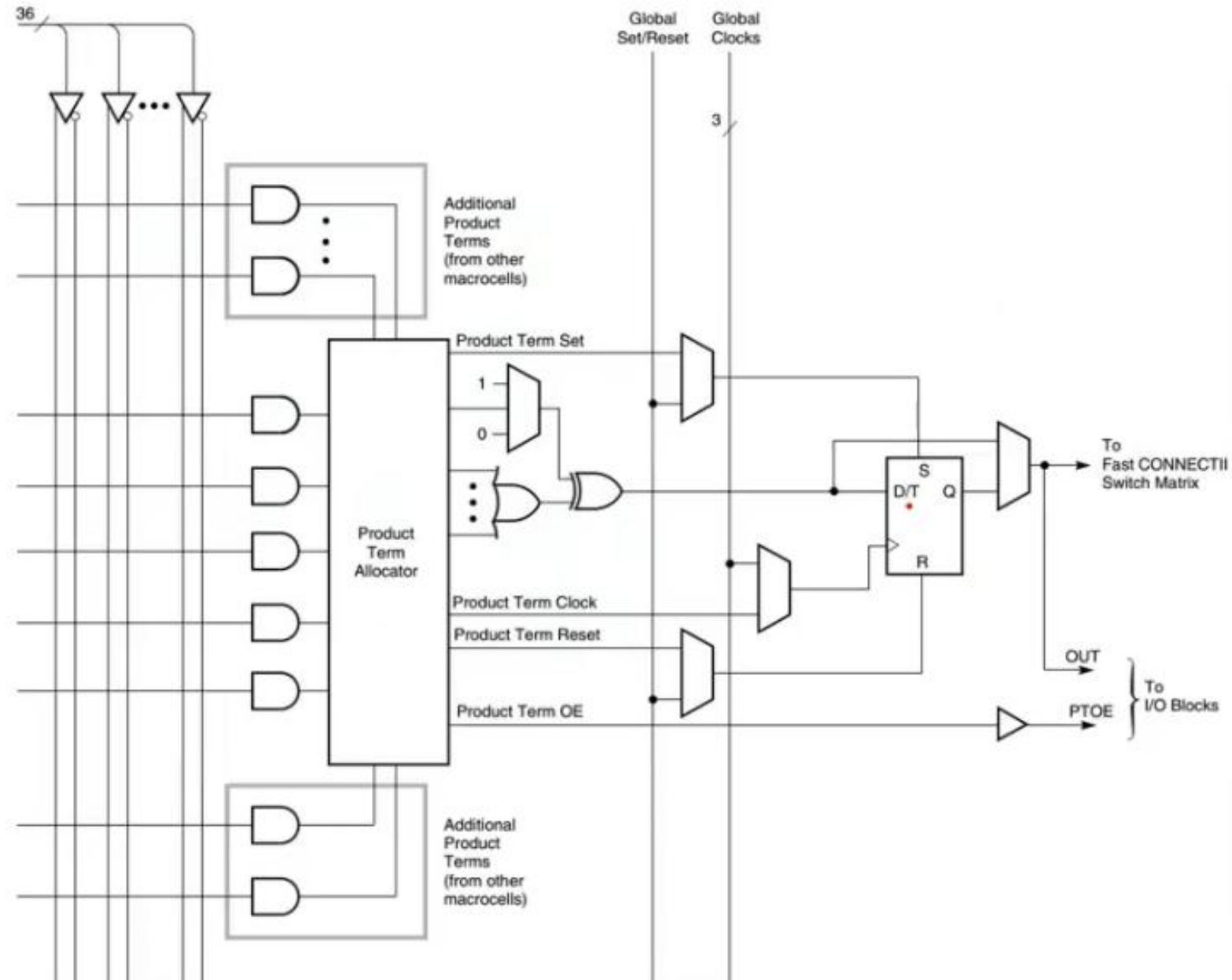


ARCHITECTURE OF XILINX 9500-FAMILY CPLD



XILINX 9500-FAMILY CPLD- FB





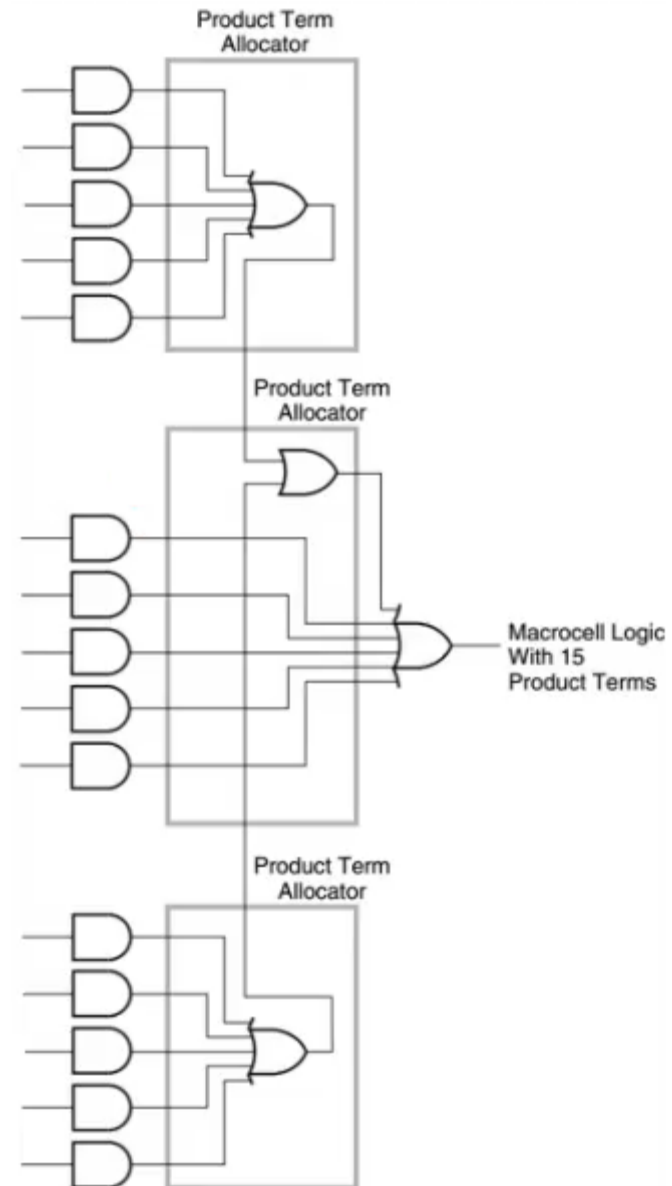
PRODUCT TERM ALLOCATOR (PTA)

- **Flexible Allocation:**

The PTA allows product terms generated in one logic block to be used by other macrocells within the same logic block or across different logic blocks.

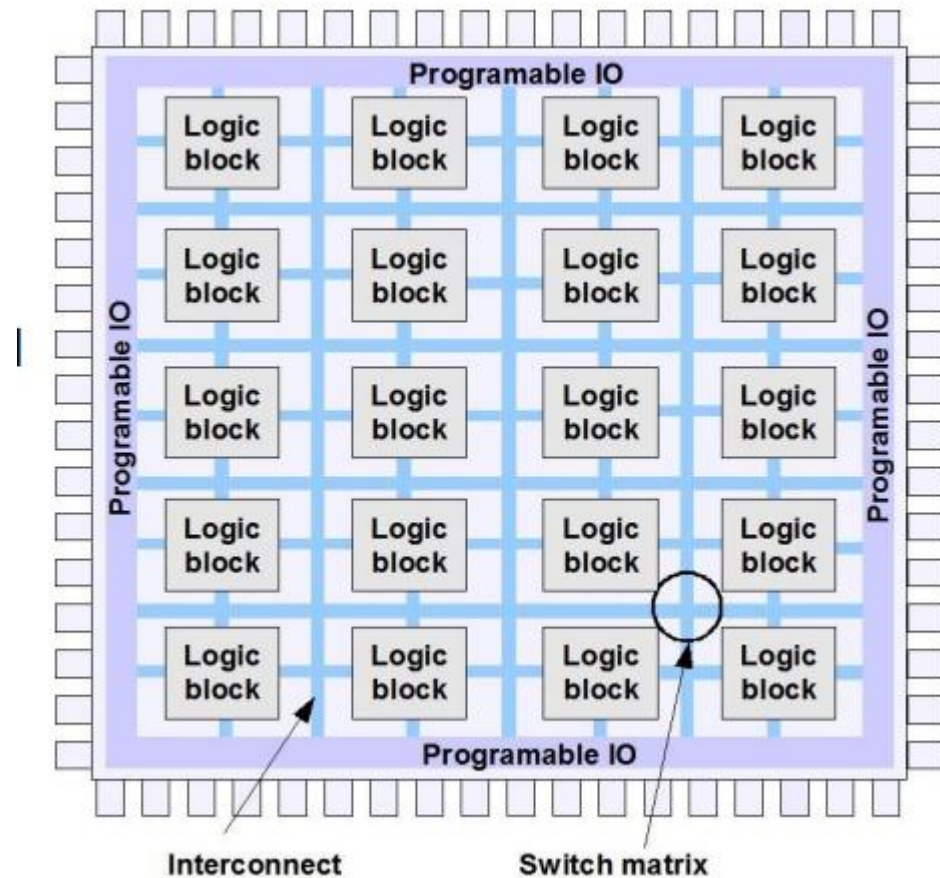


PTA



FIELD PROGRAMMABLE GATE ARRAYS (FPGAs)

Introduced in 1985 by Xilinx



ELEMENTS OF FPGA

- Combinational/Sequential Logic
- Interconnect
- IO Blocks

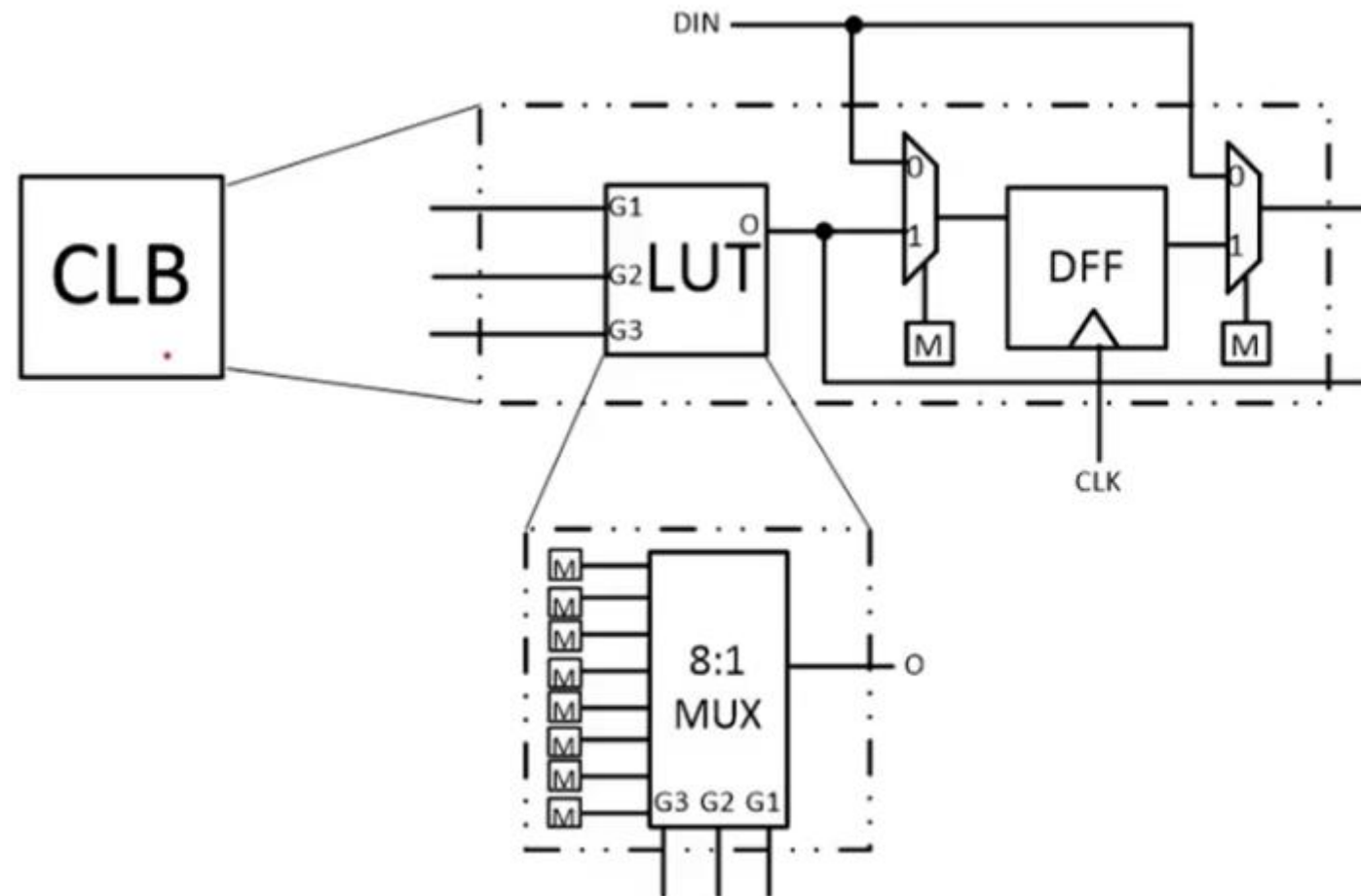


TERMINOLOGIES

- **Configuration**: bits that determine the logic function is implemented and the interconnect
- **LUT**: Look up table- Memory + Multiplexor
- **CLB/ Logic Element/Function Block/Logic Block /ALM**: Configurable Logic Block/Adaptive Logic Module
- **IO Block** : IO Pins + Associated logic



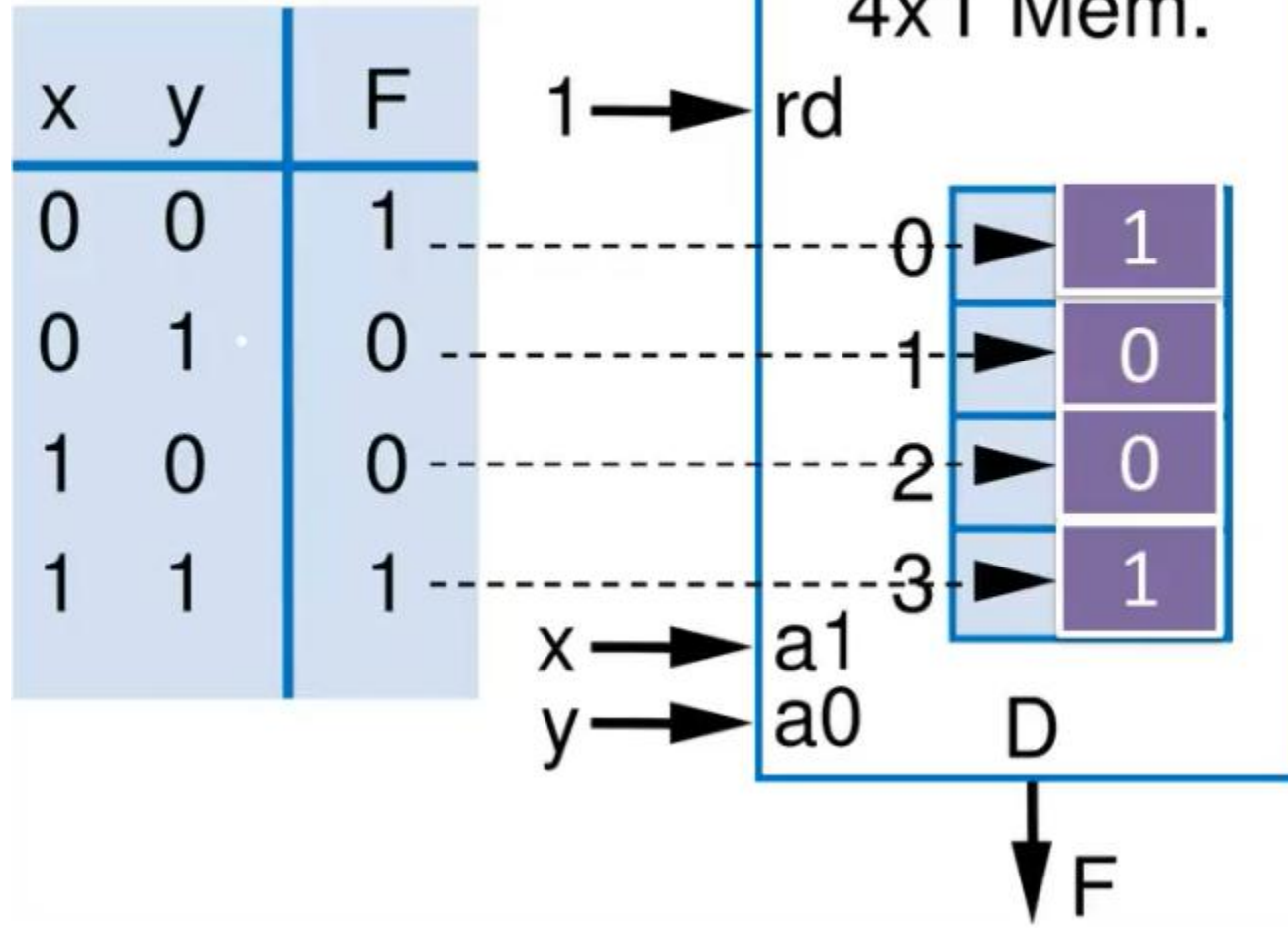
CLB ARCHITECTURE



LUT AS A MEMORY

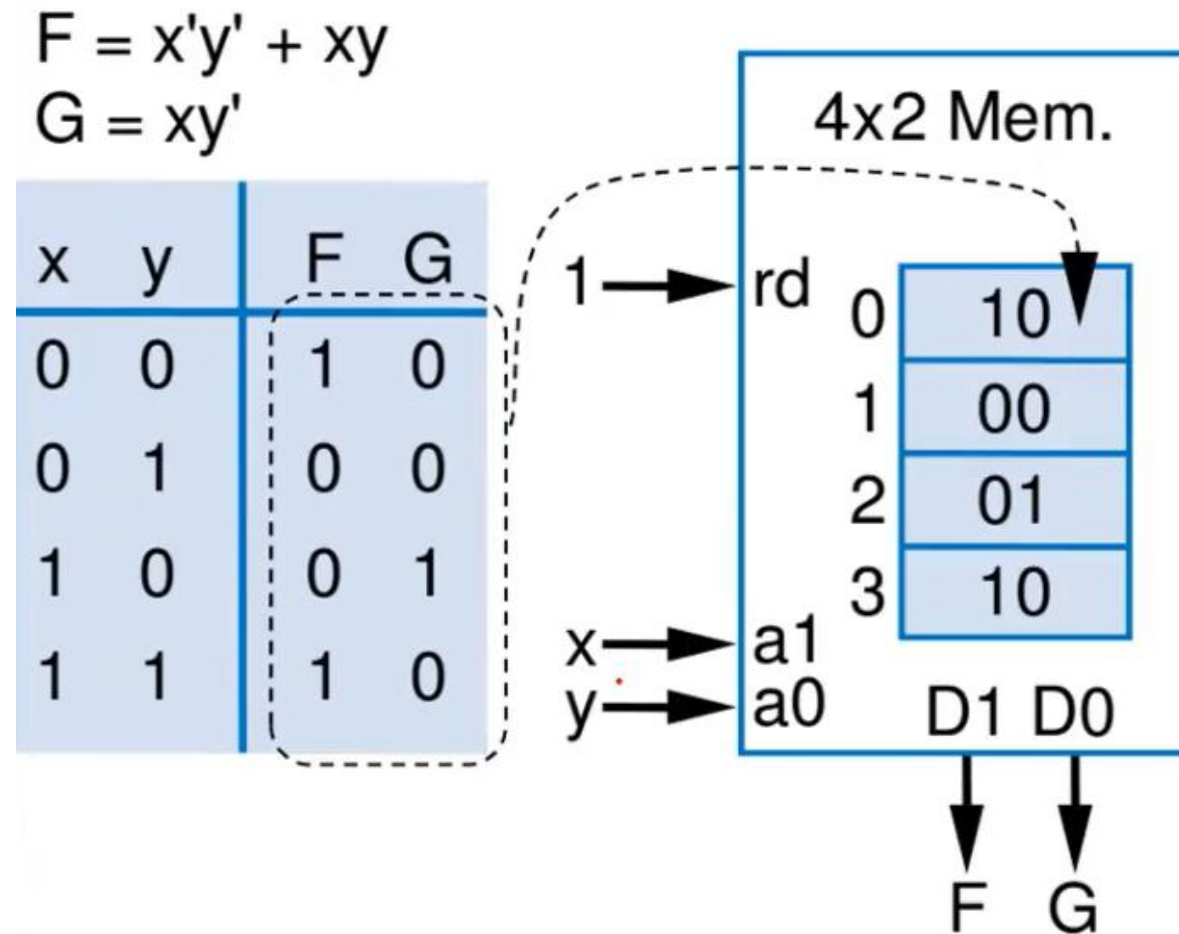
2i/p , 1 o/p LUT

$$F = x'y' + xy$$

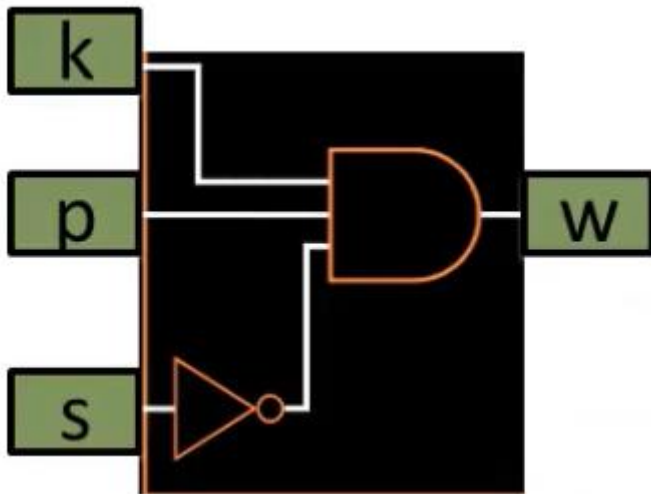


LUT AS A MEMORY

2i/p , 2 o/p LUT



EXAMPLE -LUT



k	p	s	w
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

8x1 Mem.

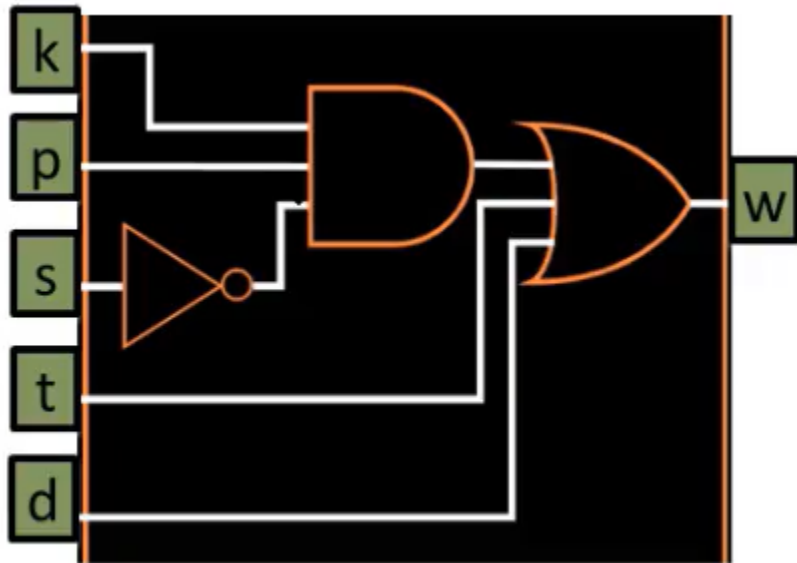
0	0
1	0
2	0
3	0
4	0
5	0
6	1
7	0

a2
a1
a0

D



EXAMPLE –IMPLEMENT 51/P CIRCUIT USING 3 I/P LUT'S



8x1 Mem.

0	0
1	0
2	0
3	0
4	0
5	0
6	1
7	0

a2
a1
a0

D

8x1 Mem.

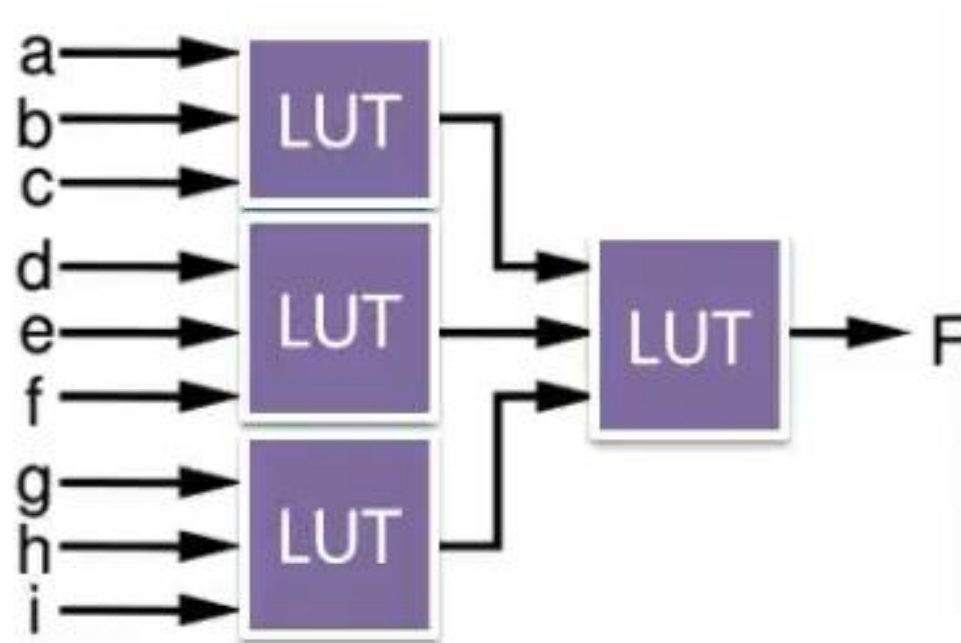
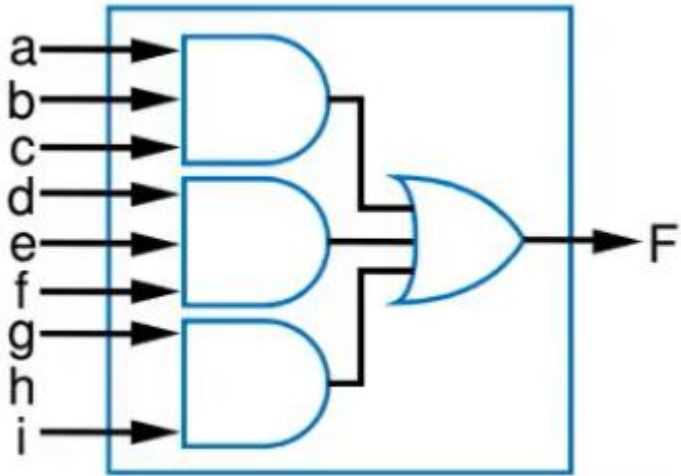
0	0
1	0
2	0
3	0
4	0
5	0
6	1
7	0

a2
a1
a0

D



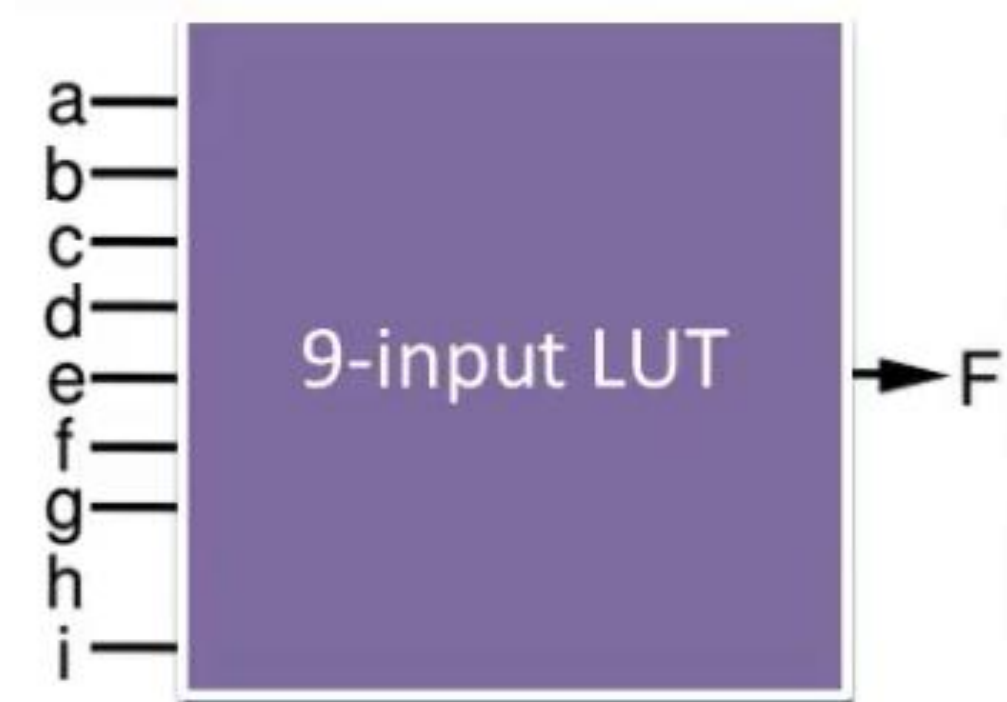
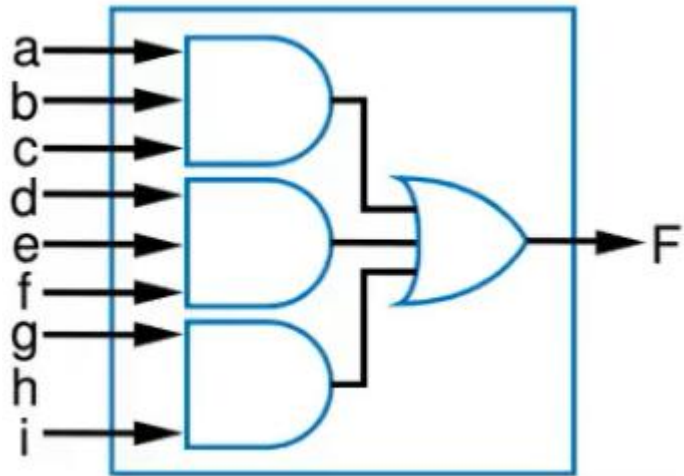
EXAMPLE –IMPLEMENT 9 I/P CIRCUIT USING 3 I/P LUT'S



4 , 8x1 Mem



ANOTHER IMPLEMENTATION...

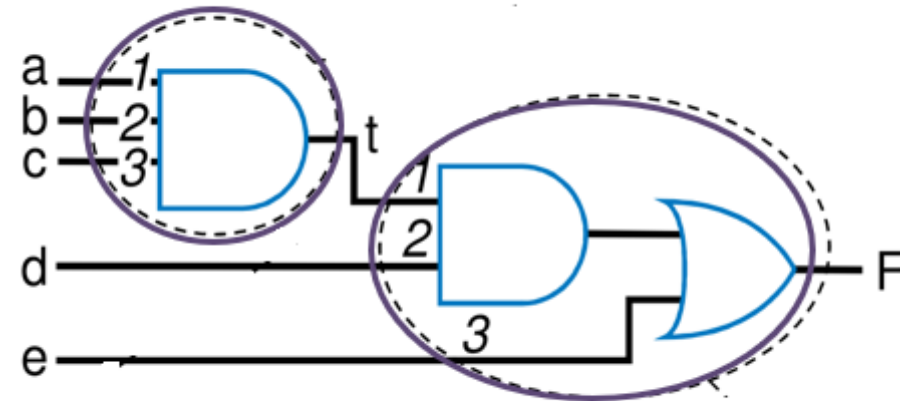
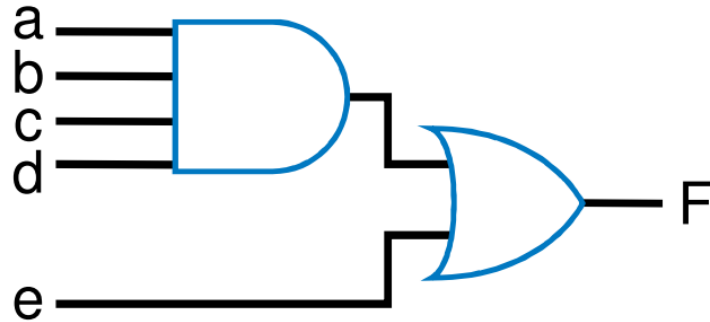


1 , 512x1 Mem

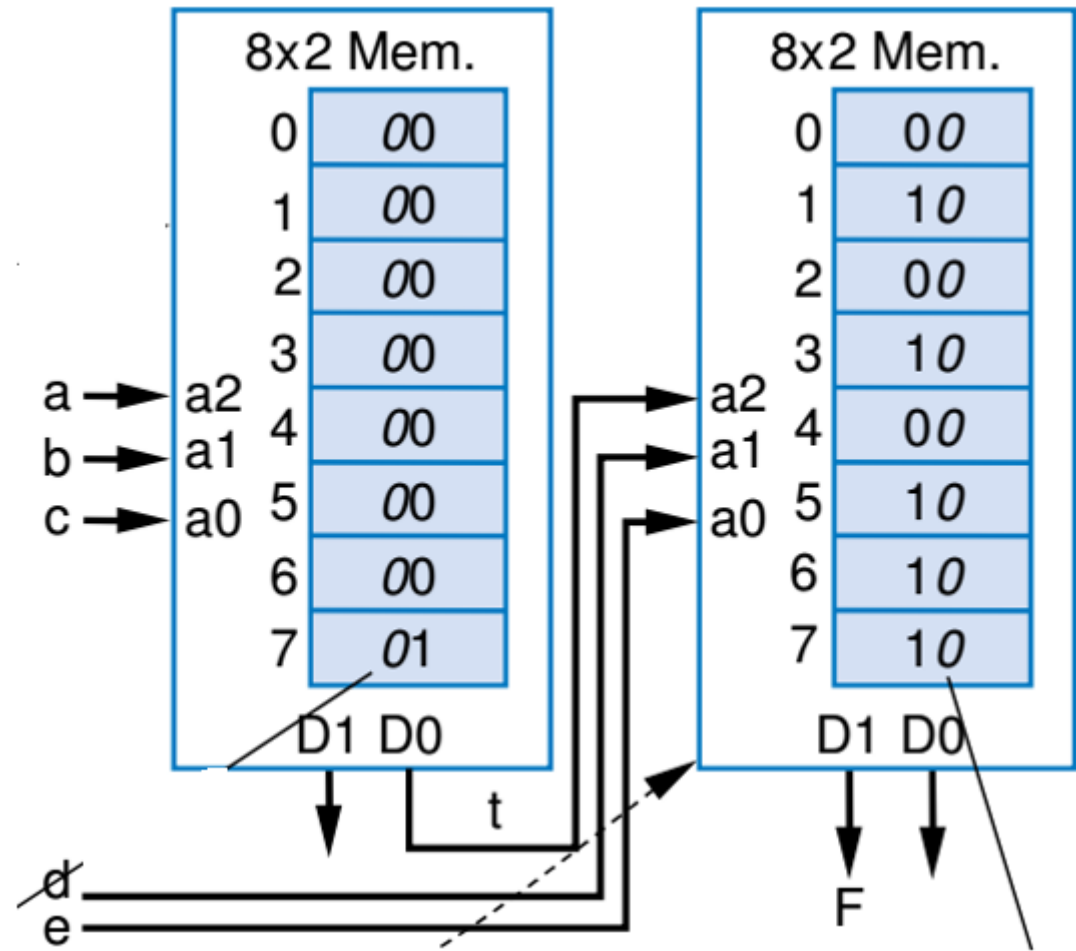
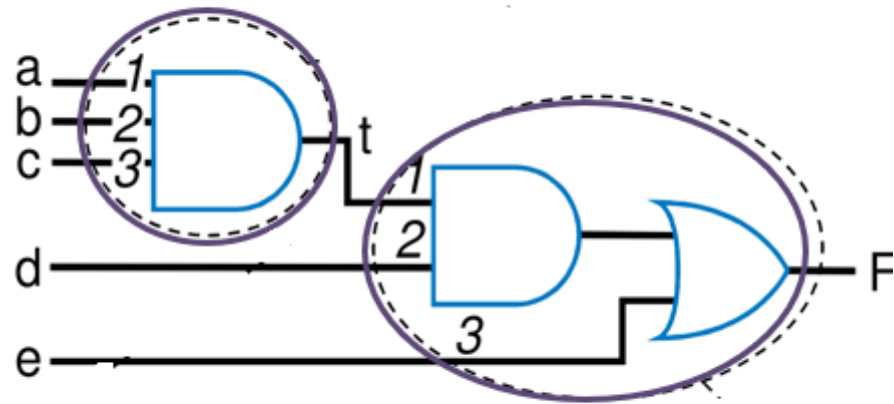


EXAMPLE

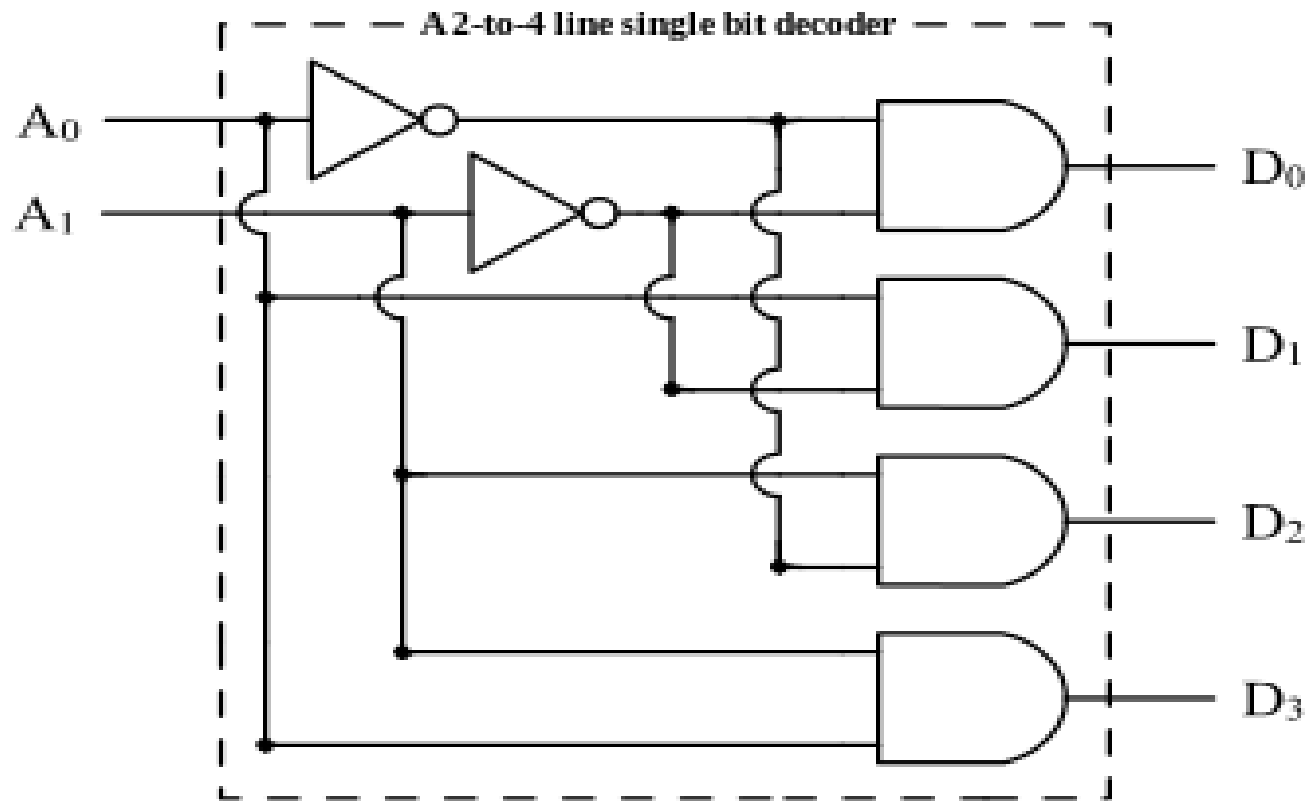
Example: Partitioning among 3-input, 2-output LUTs



EXAMPLE



IMPLEMENT 2X4 DECODER USING 3-INPUT 2-OUTPUT LUTS



Truth Table

A_1	A_0	D_3	D_2	D_1	D_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

Minterm Equations

$$D_0 = \overline{A_1} \cdot \overline{A_0}$$

$$D_1 = \overline{A_1} \cdot A_0$$

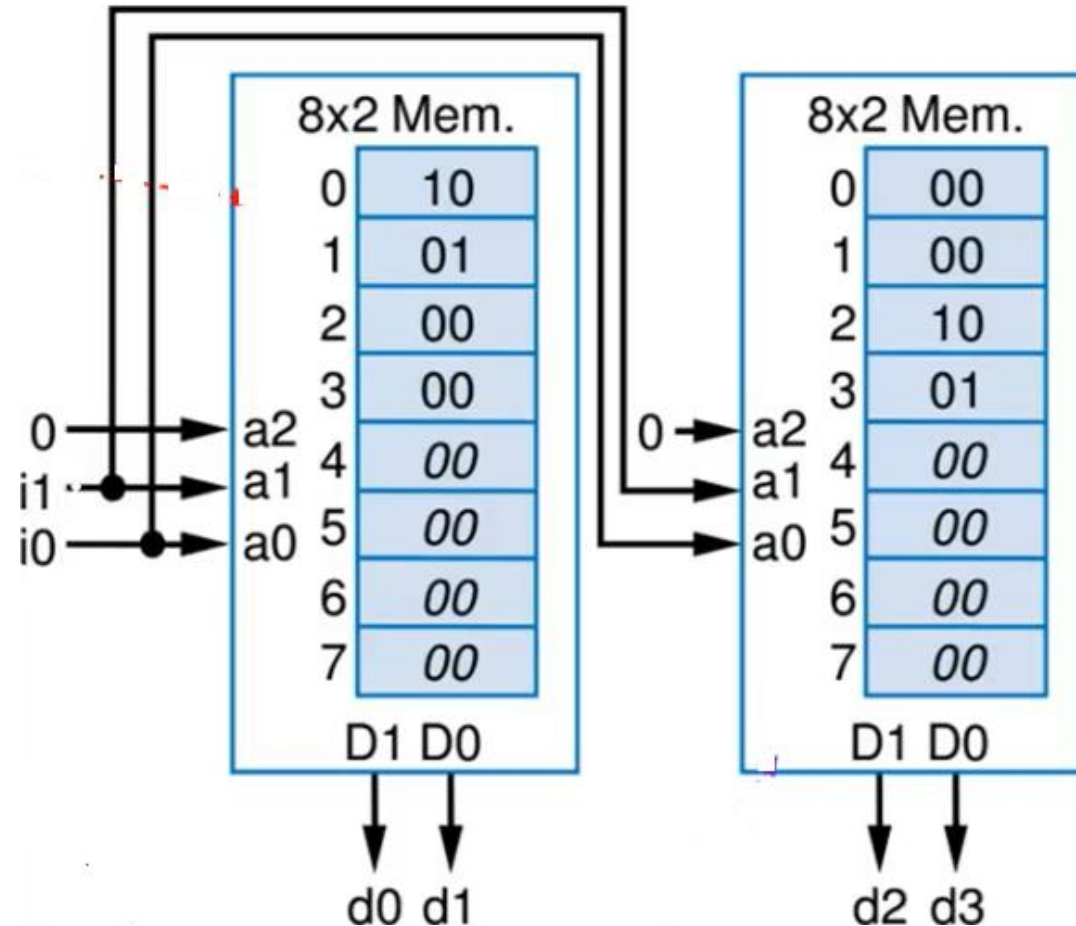
$$D_2 = A_1 \cdot \overline{A_0}$$

$$D_3 = A_1 \cdot A_0$$



IMPLEMENT 2X4 DECODER USING 3-INPUT 2-OUTPUT LUTS

Truth Table					
A_1	A_0	D_3	D_2	D_1	D_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0



LUT

